Lecture 13

If quantum mechanics really describes the world, it should also capture classical probabilistic processes. In particular, probabilistic bits should occur as a special case of qubits. However, this is not the case in our current formalism.

Density matrix

If we measure $|\psi\rangle \in \mathbb{C}^d$ in the standard basis, we get outcome $k \in \{1, \dots, d\}$ with probability

$$p_k = |\langle k|\psi\rangle|^2 = \langle k|\psi\rangle \cdot \overline{\langle k|\psi\rangle}$$

$$= \langle k|\psi\rangle \cdot (\langle k|\psi\rangle)^\dagger$$

$$= \langle k|\psi\rangle\langle\psi|k\rangle$$

$$= \langle k|\rho|k\rangle$$

where $\rho := |\psi\rangle\langle\psi|$ is called the density matrix of $|\psi\rangle$. If we were to measure $|\psi\rangle$ in any other orthonormal basis $|b_1\rangle, \dots, |b_d\rangle$, the outcome probabilities can still be obtained from ρ using a similar formula: $\langle b_k|\rho|b_k\rangle$.

Conveniently, ρ does not depend on the global phase of $|\psi\rangle$ since $(e^{i\varphi}|\psi\rangle)^\dagger = e^{-i\varphi}\langle\psi|$, so it cancels out.

Mixed states

Let $|\psi_1\rangle, \dots, |\psi_n\rangle \in \mathbb{C}^d$ be arbitrary states and let $q_1, \dots, q_n \in [0,1]$ be a probability distribution (i.e., $q_i \geq 0$ and $\sum_{i=1}^n q_i = 1$). We call the states $|\psi_i\rangle$ together with the probabilities q_i an ensemble.

What are the outcome probabilities if you pick the state $|\psi_i\rangle$ with probability q_i and measure in the standard basis? You get $k \in \{1, \dots, d\}$ with probability

$$\begin{aligned} p_k &= \sum_{i=1}^n q_i \, |\langle k|\psi_i\rangle|^2 \\ &= \sum_{i=1}^n q_i \, \langle k|\psi_i\rangle\langle\psi_i|k\rangle \\ &= \langle k| \underbrace{\left(\sum_{i=1}^n q_i \, |\psi_i\rangle\langle\psi_i| \right)}_{\rho} |k\rangle \end{aligned}$$

In this case, we can also define a density matrix that fully determines the outcome probabilities. We call the states $|\psi_i\rangle$ pure and ρ mixed because ρ is a probability distribution or "mixture" over pure states.

Bloch ball

We can generalize the Bloch sphere representation of pure states from Lecture 1 to mixed states. Recall from Homework 3 that

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \iff |\psi(\vec{r})\rangle = \frac{1}{\sqrt{2(z+1)}} \begin{pmatrix} z+1 \\ x+iy \end{pmatrix} \in \mathbb{C}^2$$

establishes a correspondence between unit vectors in $\mathbb{R}^3$ (i.e., $x^2+y^2+z^2=1$) and unit vectors in $\mathbb{C}^2$ with the global phase ignored (also known as the complex projective space).

You showed in Homework 3 that the density matrix of $|\psi(\vec{r})\rangle$ is given by

$$\rho(\vec{r}) = |\psi(\vec{r})\rangle\langle\psi(\vec{r})| = \frac{1}{2}(I + xX + yY + zZ)$$

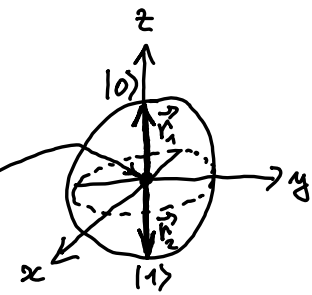
which is much easier to remember than the formula for $|\psi(\vec{r})\rangle$. We call $\vec{r}$ the <u>Bloch vector</u> of $|\psi(\vec{r})\rangle$ or $\rho(\vec{r})$.

If you pick $|\psi(\vec{r}_1)\rangle$ with probability q_1 and $|\psi(\vec{r}_2)\rangle$ with probability q_2, the resulting density matrix is

$$\begin{aligned}
&q_1 \rho(\vec{r}_1) + q_2 \rho(\vec{r}_2) \\
&= q_1 \frac{1}{2}(I + x_1 X + y_1 Y + z_1 Z) \\
&+ q_2 \frac{1}{2}(I + x_2 X + y_2 Y + z_2 Z) \\
&= \frac{1}{2}\Big(I + (q_1 x_1 + q_2 x_2) X \\
&\qquad + (q_1 y_1 + q_2 y_2) Y \\
&\qquad + (q_1 z_1 + q_2 z_2) Z\Big) \\
&= \rho(q_1 \vec{r}_1 + q_2 \vec{r}_2)
\end{aligned}$$

For example, if $\vec{r}_1 = \begin{pmatrix}0\\0\\1\end{pmatrix}$, $\vec{r}_2 = \begin{pmatrix}0\\0\\-1\end{pmatrix}$, and $q_1 = q_2 = \frac{1}{2}$ then

$$q_1 \vec{r}_1 + q_2 \vec{r}_2 = \frac{1}{2}\begin{pmatrix}0\\0\\1-1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}.$$



The resulting state is called the <u>maximally mixed</u> state:

$$\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix}1&0\\0&1\end{pmatrix} = \frac{1}{2}I.$$

Note that it can be obtained from other states as well:

$$\frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = \frac{1}{2}\begin{pmatrix}1&0\\0&1\end{pmatrix}.$$

Note that $|+\rangle \Rightarrow \vec{r}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}$ and $|-\rangle \Rightarrow \vec{r}_2 = \begin{pmatrix}-1\\0\\0\end{pmatrix}$, so

$\frac{1}{2}r_1 + \frac{1}{2}r_2 = \frac{1}{2}\begin{pmatrix}1-1\\0\\0\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$ is also at the origin.

In fact,

$$\frac{1}{2}\rho(\vec{r}) + \frac{1}{2}\rho(-\vec{r}) = \frac{1}{2}I$$

for any $\vec{r} \in \mathbb{R}^3$. Recall from Homework 3 that $\{|\psi(\vec{r})\rangle, |\psi(-\vec{r})\rangle\}$ is an orthonormal basis, for any unit vector $\vec{r}$.

Any $\vec{r} \in \mathbb{R}^3$ such that $\|\vec{r}\| \leq 1$ describes a valid quantum state. Together they form the Bloch ball.

Intuitively, the length $\|\vec{r}\|$ of the Bloch vector measures the "purity" of the state $\rho(\vec{r})$.

- $\|\vec{r}\| = 1 \Rightarrow$ pure
- $\|\vec{r}\| < 1 \Rightarrow$ mixed
- $\|\vec{r}\| = 0 \Rightarrow$ maximally mixed



The eigenvalues of $\rho(\vec{r})$ are $\frac{1}{2}(1 \pm \|\vec{r}\|)$, which are $\{0,1\}$ for pure states and $\{\frac{1}{2},\frac{1}{2}\}$ for max. mixed.

Mixed states of dimension d

When does a matrix $\rho \in M_d(\mathbb{C})$ describe a valid mixed state?

- ρ is Hermitian: $\left(\sum_{i=1}^{n} q_i |\psi_i\rangle\langle\psi_i|\right)^+ = \sum_{i=1}^{n} q_i |\psi_i\rangle\langle\psi_i|$.
- Measurement probabilities in any basis are non-negative:
$$\langle\varphi|\rho|\varphi\rangle \geq 0 \quad \forall |\varphi\rangle \in \mathbb{C}^d$$
- The probabilities (when measuring in the standard basis) sum to 1:
$$\sum_{k=1}^{d} \langle k|\rho|k\rangle = 1$$

Matrices obeying the first two properties are called <u>positive semidefinite</u>. This constraint is abbreviated as $\rho \geq 0$. (This does not mean ρ is entry-wise non-negative.) This is equivalent to ρ being Hermitian, $\rho^+ = \rho$, and having non-negative eigenvalues.

The third condition is abbreviated as $\mathrm{Tr}\,\rho = 1$ where
$$\mathrm{Tr}\,\rho := \sum_{k=1}^{d} \rho_{kk}$$
is called the <u>trace</u> of ρ. It's just the sum of the diagonal entries.

You can think of mixed states as "non-commutative" probability distributions. Diagonal states represent actual prob. distributions:
$$\rho = \begin{pmatrix} p_1 & & 0 \\ & p_2 & \\ & \ddots & \\ 0 & & p_d \end{pmatrix}$$

Evolution of mixed states

A pure state $|\psi\rangle$ evolves under a unitary operation U as follows:

$$|\psi\rangle \mapsto U|\psi\rangle.$$

Since $(U|\psi\rangle)^\dagger = \langle\psi|U^\dagger$, the corresponding density matrix evolves as

$$|\psi\rangle\langle\psi| \mapsto U|\psi\rangle\langle\psi|U^\dagger.$$

In fact, any mixed state ρ evolves as

$$\rho \mapsto U\rho U^\dagger.$$

In addition to unitary evolution, we can also add a new "clean" register in state $|0\rangle$:

$$|\psi\rangle \mapsto |\psi\rangle \otimes |0\rangle \quad \text{or} \quad \rho \mapsto \rho \otimes |0\rangle\langle 0|.$$

This is not a unitary operation since it increases the dimension.

We can also discard a register. Given some state $|\Psi\rangle_{AB}$, Alice can discard her system A. What is Bob's remaining state?

One way to think about this is to imagine that Alice measures her register in the standard basis and forgets the measurement outcome. We can then compute Bob's reduced state by averaging over all measurement outcomes.

Reduced state

Let $|\Psi\rangle_{AB} \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ be an arbitrary pure state shared between Alice and Bob. What is Bob's state if Alice measures her register and forgets the outcome? She gets outcome $k \in \{1, \ldots, d_A\}$ with probability

$$p_k = \| (\langle k|_A \otimes I_B) |\Psi\rangle_{AB} \|^2$$

and Bob's remaining state in this case is

$$|\psi_k\rangle_B = \frac{(\langle k|_A \otimes I_B)|\Psi\rangle_{AB}}{\sqrt{p_k}}.$$

If Alice forgets her measurement outcome k, the average state on Bob's system is

$$\sum_{k=1}^{d_A} p_k |\psi_k\rangle\langle\psi_k| = \sum_{k=1}^{d_A} (\langle k|_A \otimes I_B)|\Psi\rangle\langle\Psi|_{AB}(|k\rangle_A \otimes I_B)$$
$$= \mathrm{Tr}_A(|\Psi\rangle\langle\Psi|_{AB}) = \rho_B$$

where Tr_A denotes the <u>partial trace</u> over Alice's system. In general, Bob's <u>reduced state</u> is found by computing the partial trace of their shared state:

$$\rho_B = \mathrm{Tr}_A(\rho_{AB}) := \sum_{k=1}^{d_A} (\langle k|_A \otimes I_B)\, \rho_{AB}\, (|k\rangle_A \otimes I_B).$$

Note how this is similar to computing the regular trace, $\mathrm{Tr}\,\rho_A = \sum_{k=1}^{d_A} \langle k|\rho_A|k\rangle$, except we also do nothing on the B register. The partial trace of B register is defined similarly.

Example (a product state) Let $|\Psi\rangle_{AB} = |\alpha\rangle_A \otimes |\beta\rangle_B$. Note that

$$|\Psi\rangle\langle\Psi|_{AB} = (|\alpha\rangle_A \otimes |\beta\rangle_B)(\langle\alpha|_A \otimes \langle\beta|_B)$$

$$= |\alpha\rangle\langle\alpha|_A \otimes |\beta\rangle\langle\beta|_B$$

so

$$\mathrm{Tr}_A(|\Psi\rangle\langle\Psi|_{AB})$$

$$= \sum_{k=1}^{d_A} (\langle k|_A \otimes I_B)(|\alpha\rangle\langle\alpha|_A \otimes |\beta\rangle\langle\beta|_B)(|k\rangle_A \otimes I_B)$$

$$= \sum_{k=1}^{d_A} \langle k|\alpha\rangle\langle\alpha|k\rangle \otimes |\beta\rangle\langle\beta|_B$$

$$= \underbrace{\left(\sum_{k=1}^{d_A} |\langle k|\alpha\rangle|^2\right)}_{1} |\beta\rangle\langle\beta|_B = |\beta\rangle\langle\beta|_B$$

Since Tr_A corresponds to throwing away the A system, this is indeed the result we would expect.

Example (max. entangled state) What remains if we throw away the A system of

$$|\beta_{00}\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \; ?$$

Its density matrix is $|\beta_{00}\rangle\langle\beta_{00}|_{AB} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$ and the remaining state

$$\mathrm{Tr}_A(|\beta_{00}\rangle\langle\beta_{00}|_{AB}) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is maximally mixed. There is a sense in which the amount of entanglement in a pure bipartite state can be measured by the mixedness or "entropy" of the reduced state on one of the systems.

General quantum operation

The most general quantum operation on quantum channel consists of the following steps:

1. Add a new register initialized in some fixed state, say $|0\rangle$.
2. Apply a unitary on both registers.
3. Discard the second register.

This is known as a Stinespring representation of a quantum channel. Note that this includes adding a new register, applying a unitary on the given state, or discarding a register as special cases.

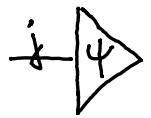
If you want to learn more about mixed states, quantum channels and entanglement, you should take Quantum Information Theory during your master's.

Graphical notation

There is an intuitive way of performing certain types of calculations that involves pictures instead of equations. This graphical notation is also known as tensor networks.

- Column vector $|\psi\rangle \in \mathbb{C}^d$

$$|\psi\rangle = \sum_{j=1}^{d} \psi_j |j\rangle$$



- Matrix $A \in M_d(\mathbb{C})$

$$A = \sum_{i,j=1}^{d} a_{ij} |i\rangle\langle j|$$



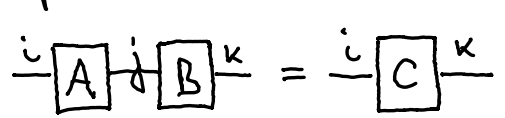
- Matrix – vector multiplication:

$$\forall i: \sum_{j=1}^{d} A_{ij} \psi_j = \varphi_i$$



- Matrix – matrix multiplication:

$$\forall i,k: \sum_{j=1}^{d} A_{ij} B_{jk} = C_{ik}$$

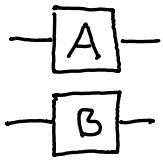


- Identity matrix:

$$I = \sum_{i=1}^{d} |i\rangle\langle i|$$

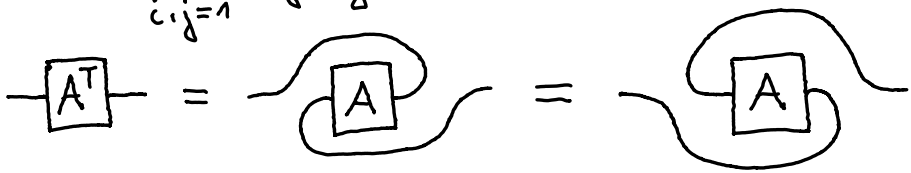
- Tensor product:

$$A \otimes B$$



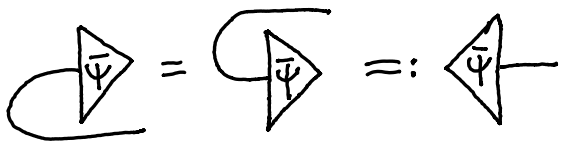
- Matrix transpose:

$$A^T = \sum_{i,j=1}^{d} a_{ij} |j\rangle\langle i|$$



- Vector (conjugate) transpose:

$$\langle\psi| = |\psi\rangle^T =$$



$$|\psi\rangle =$$

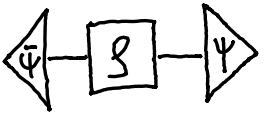


$$\langle\psi| =$$



- Probability to get $|\psi\rangle$ when measuring ρ:

$$\langle\psi|\rho|\psi\rangle =$$

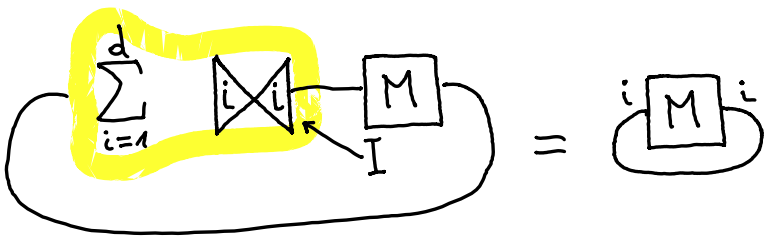


- Trace:

$$\mathrm{Tr}\, M = \sum_{i=1}^{d} \langle i|M|i\rangle = \sum_{i=1}^{d}$$

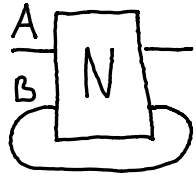


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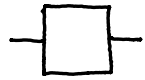


- Partial trace:

$$\mathrm{Tr}_B N_{AB} =$$



$$=$$

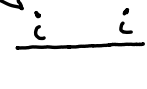


- Un-normalized maximally entangled state:

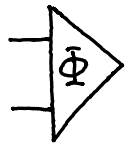
$$I = \sum_{i=1}^{d} |i\rangle\langle i| = \sum_{i=1}^{d}$$



$$=$$



$$|\Phi\rangle := \sum_{i=1}^{d} |i\rangle|i\rangle =$$



$$=$$

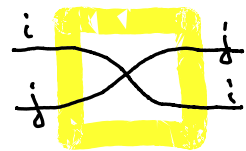


$$\langle\Phi| =$$



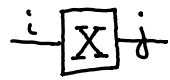
- SWAP gate:

$$\mathrm{SWAP} = \sum_{i,j=1}^{d} |i\rangle|j\rangle\langle j|\langle i| =$$

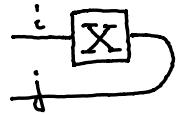


- Vectorization:

$$X = \sum_{i,j=1}^{d} x_{ij} |i\rangle\langle j| =$$

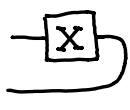


$$\mathrm{vec}(X) = \sum_{i,j=1}^{d} x_{ij} |i\rangle|j\rangle =$$

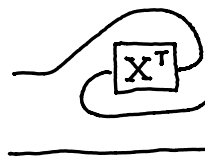


- Sliding:

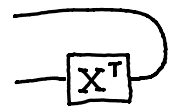
$$\mathrm{vec}(X) =$$



$$=$$

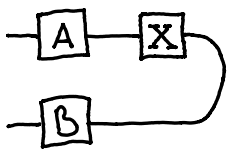


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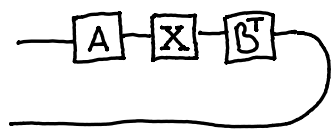


- Vectorization identity:

$$(A \otimes B)\,\mathrm{vec}(X) = \mathrm{vec}(AXB^T)$$



$$=$$



Purification

Given a mixed state ρ_A, find a pure state $|\Psi\rangle_{AB}$ such that discarding the B system produces ρ_A:

$$\mathrm{Tr}_B(|\Psi\rangle\langle\Psi|_{AB}) = \rho_A$$

Such $|\Psi\rangle_{AB}$ is known as a purification of ρ_A.

Ex. The maximally entangled state

$$|\beta\rangle_{AB} = \frac{1}{\sqrt{d}}|\Phi\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^{d}|i\rangle|i\rangle$$

is a purification of the maximally mixed state I/d since

$$\mathrm{Tr}_B(|\beta\rangle\langle\beta|_{AB}) = \frac{1}{d}$$



$$= \frac{1}{d}\ \text{—}\ = \frac{1}{d}I$$

For a general state ρ, with spectral decomposition

$$\rho = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|, \quad \lambda_i \geq 0,$$

its operator square root is given by

$$\sqrt{\rho} = \sum_i \sqrt{\lambda_i}|\psi_i\rangle\langle\psi_i|.$$

You can verify that $\sqrt{\rho}\sqrt{\rho} = \rho$ and

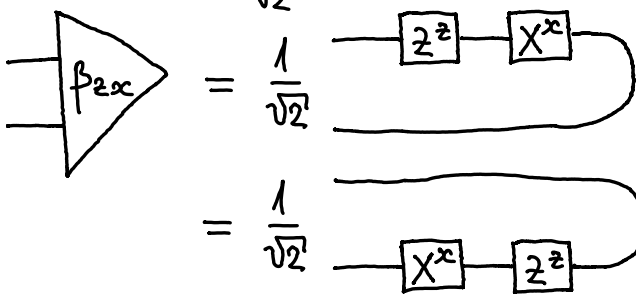
$$|\Psi\rangle_{AB} := (\sqrt{\rho}_A \otimes I_B)|\Phi\rangle_{AB}$$

is a purification of ρ_A.

Bell states and Pauli matrices

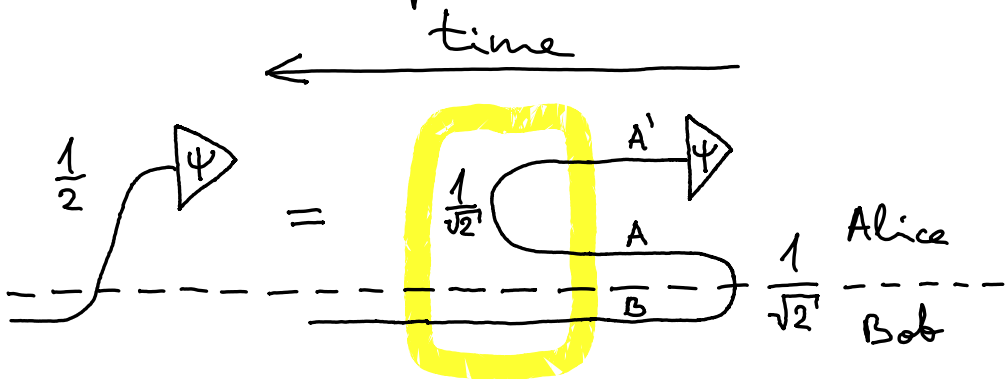
Bell states are related to Pauli matrices as follows (local conversion identities):

$$|\beta_{zx}\rangle = \frac{1}{\sqrt{2}} \operatorname{vec}(Z^z X^x) \qquad \forall z, x \in \{0,1\}$$



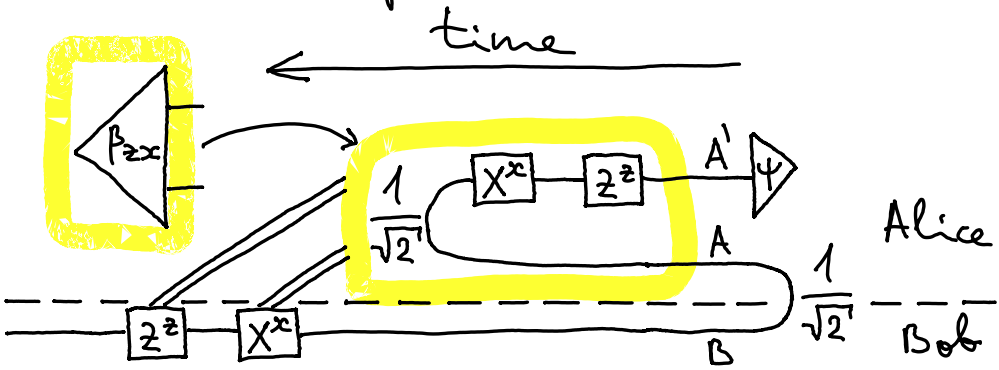
Teleportation

Here is how you can derive and analyze the teleportation protocol based on diagrams. Alice has an unknown state $|\psi\rangle_{A'} \in \mathbb{C}^2$ and shares with Bob the EPR state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{AB} = \frac{1}{\sqrt{2}}|\Phi\rangle_{AB}$. If she measures her systems $A'A$ in the Bell basis and gets the outcome 00 corresponding to the EPR state $|\beta_{00}\rangle = \frac{1}{\sqrt{2}}|\Phi\rangle$, Bob's state is $|\psi\rangle_B$:

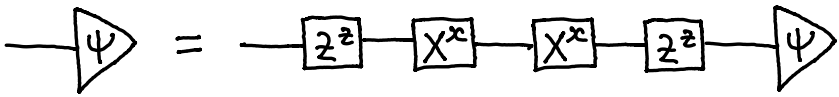


This happens with probability $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

More generally, if Alice gets outcomes $z, x \in \{0,1\}$, she sends them to Bob who needs to apply the Pauli correction operator $Z^z X^x$:



This reproduces the correct state $|\psi\rangle$ since



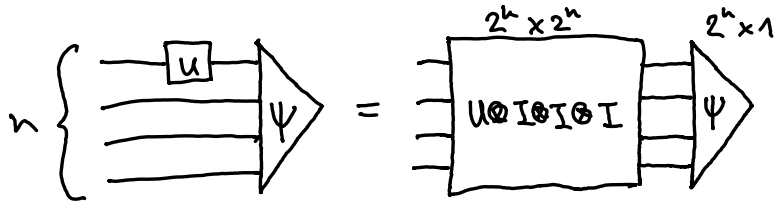
because $X^0 = X^2 = I = Z^0 = Z^2$.

Graphical notation and tensor networks is not only a fun theoretical tool but also has important practical applications. One of such applications is the simulation of quantum circuits on a classical computers, such as in Cirq.

Simulation of quantum circuits: naive approach vs tensor networks

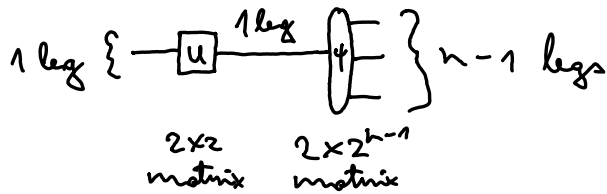
Let's say we store an n-qubit state $|\psi\rangle$ as a vector with 2^n components. How expensive is it to apply a 1-qubit gate on the first qubit?

Naive approach – treat U as a $2^n \times 2^n$ matrix:



This has complexity $2^n \cdot 2^n \cdot 1 = 2^{2n}$.

On the other hand, if we treat U and $|\psi\rangle$ as tensors and only contract one index, we are effectively multiplying smaller matrices:



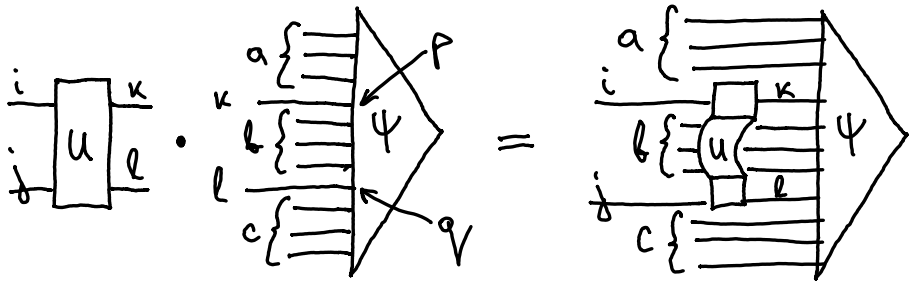
This has complexity $2 \cdot 2 \cdot 2^{n-1} = 2^{n+1}$, which is much better than the naive approach. Hence this is also the method used by many simulators, such as the one in Cirq.

Let's take a look at how to implement this in Python.

Example Python code

If we store an n-qubit state $|\psi\rangle$ as a vector with 2^n components, we can apply 1-qubit and 2-qubit gates by contracting the legs on which the gate acts with a tensor describing the gate itself.

Let's say we want to apply a 2-qubit gate U on qubits $p<q$ of $|\psi\rangle$:



Create all-zeroes vector with 2^n complex entries and initialize to $|0\rangle^{\otimes n}$:

```
state = np.zeros(2**n, np.complex64)
state[0] = 1+0j
```

Group legs into 5 groups of sizes $p, 1, q-p-1, 1, n-q-1$:

```
psi = state.reshape([2**p, 2, 2**(q-p-1), 2, 2**(n-q-1)])
```

Contract the gate and state tensors:

```
state = np.einsum('ijkl, akblc->aibjc', U, psi)
```

This is essentially what happens inside Cirq.